

Week 7

4H-SiC LGAD TCAD | Week 7 update

# SICAR geometry replication

Replicating the SICAR 4H-SiC LGAD geometry in Victory Device

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## **A. SICAR reference geometry**

Layer stack, SIMS-validated doping, the C-V target signature at 65–80 V.

## **B. Replicated 2D geometry in Victory Device**

$48\text{ }\mu\text{m} \times 71.3\text{ }\mu\text{m}$  deck, 10-step bevel, layer-by-layer match against SICAR.

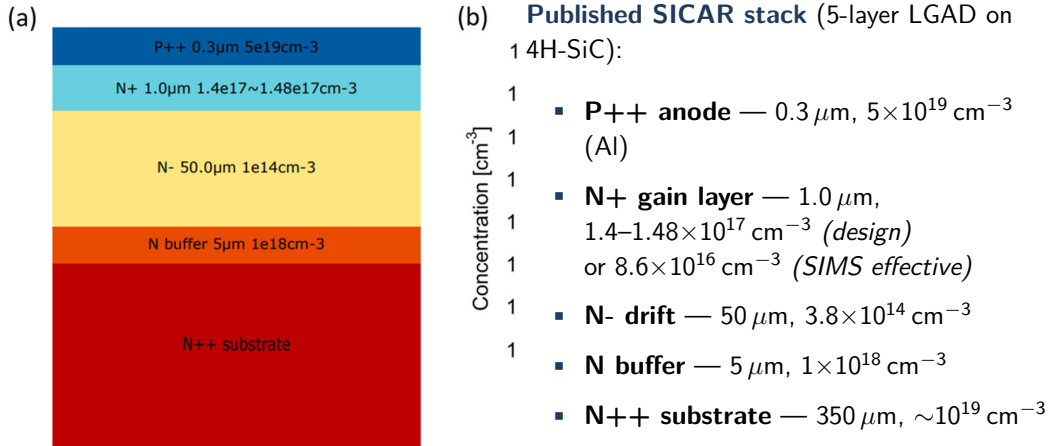
## **C. C-V diagnostics — the physics traps that block a clean C-V**

II ghost conductance, Robin boundary injection, 2D→3D capacitance scaling.

## **D. Next steps**

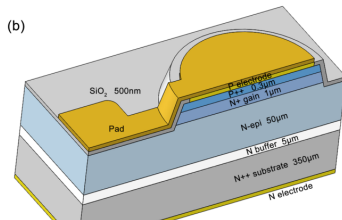
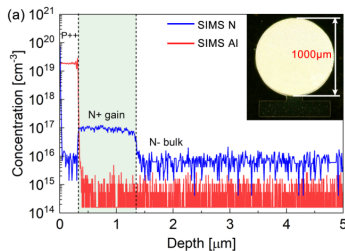
Fix the solver, extract I-V and C-V, add traps, validate with SICAR.

# SICAR reference: layer stack



SICAR timing-device paper, Fig. 1(a).

# SICAR SIMS-validated doping



## SIMS-measured values (electrical paper):

- Al peak in P++ anode:  $\sim 10^{19} \text{ cm}^{-3}$
- N peak in gain layer:  $8.6 \times 10^{16} \text{ cm}^{-3}$  (not the  $1.4 \times 10^{17}$  design target)
- N in drift:  $3.8 \times 10^{14} \text{ cm}^{-3}$

## Why SIMS, not the design value:

The fabricated device has **less gain-layer charge** than the blueprint. The C-V depletion knee shifts to a lower voltage than the design predicts.

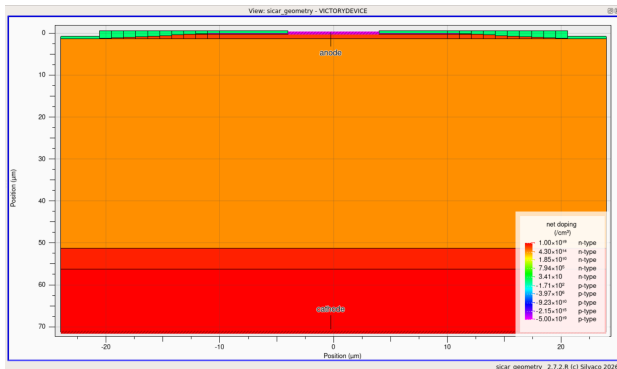
Active area =  $0.79 \text{ mm}^2$ .

SICAR electrical paper, Fig. 1: SIMS (a) + 3D schematic

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(b).

# Replicated 2D geometry — full view



## Deck dimensions:

- Width:  $48 \mu\text{m}$  ( $x: -24$  to  $+24$ )
- Height:  $71.3 \mu\text{m}$  ( $y: -0.5$  to  $71.3$ )
- 10-step bevel** on **both** edges
- $\text{SiO}_2$  passivation on bevel tops
- Anode:  $x = \pm 4 \mu\text{m}$ ,  $y = 0$
- Cathode: full back contact,  $y = 71.3$

## Substrate truncation:

$350 \mu\text{m} \rightarrow 15 \mu\text{m}$  for speed. The drift fully depletes before the substrate, so C-V is unaffected.

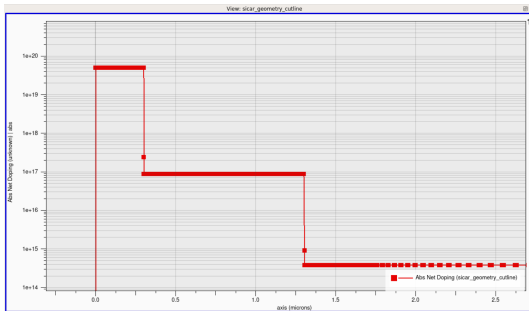
Victory Device — sicar\_geometry.str, net doping map.

## Layer-by-layer match: SICAR vs. my deck

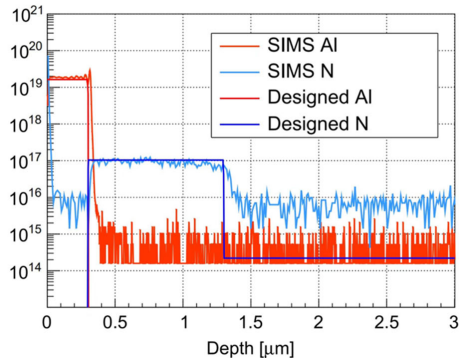
| Layer         | SICAR (paper)                                                    | My deck                                                | Source      |
|---------------|------------------------------------------------------------------|--------------------------------------------------------|-------------|
| P++ anode     | $0.3\ \mu\text{m}, 5 \times 10^{19}\ \text{cm}^{-3}$             | $0.3\ \mu\text{m}, 5 \times 10^{19}\ \text{cm}^{-3}$   | match       |
| N+ gain       | $1.0\ \mu\text{m}, 8.6 \times 10^{16}\ \text{cm}^{-3}$<br>(SIMS) | $1.0\ \mu\text{m}, 8.6 \times 10^{16}\ \text{cm}^{-3}$ | <b>SIMS</b> |
| N- drift      | $50\ \mu\text{m}, 3.8 \times 10^{14}\ \text{cm}^{-3}$            | $50\ \mu\text{m}, 3.8 \times 10^{14}\ \text{cm}^{-3}$  | match       |
| N buffer      | $5\ \mu\text{m}, 1 \times 10^{18}\ \text{cm}^{-3}$               | $5\ \mu\text{m}, 1 \times 10^{18}\ \text{cm}^{-3}$     | match       |
| N++ substrate | $350\ \mu\text{m}, \sim 10^{19}\ \text{cm}^{-3}$                 | $15\ \mu\text{m}, 1 \times 10^{19}\ \text{cm}^{-3}$    | truncated   |

The geometry is fully replicated. The remaining slides diagnose why the C-V curve is hard to extract even with a correct geometry.

# Junction detail: my deck vs. SICAR SIMS



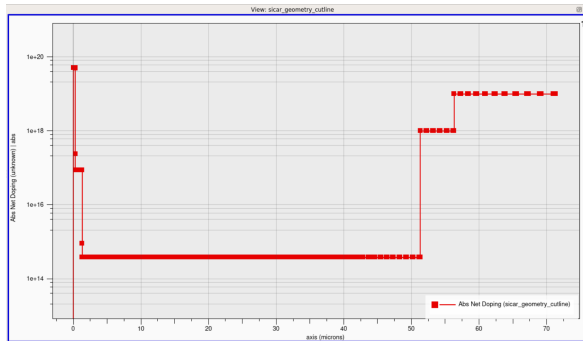
My deck — zoomed cutline, 0–2.75  $\mu\text{m}$ .



SICAR SIMS — AI & N vs. depth.

Both plots show the same junction structure: P++ anode ( $10^{19}$ )  $\rightarrow$  N+ gain ( $8.6 \times 10^{16}$ )  $\rightarrow$  N- drift ( $3.8 \times 10^{14}$ ). Transitions are **abrupt** — matching the SICAR SIMS profile at the epitaxial boundaries

# Full vertical cutline — all five layers



My deck — vertical cutline, 0–70  $\mu\text{m}$ .

## Five plateaus:

- P++ anode:  $10^{19} \text{ cm}^{-3}$
- N+ gain:  $8.6 \times 10^{16} \text{ cm}^{-3}$
- N- drift:  $3.8 \times 10^{14} \text{ cm}^{-3}$
- N buffer:  $10^{18} \text{ cm}^{-3}$
- N++ substrate:  $10^{19} \text{ cm}^{-3}$

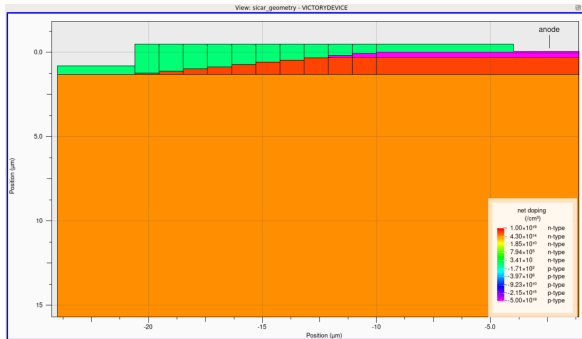
## Mesh at junctions:

- P++/N+ ( $y = 0.3$ ): 1 nm
- N+/N- ( $y = 1.3$ ): 5 nm

1 nm at P++/N+ keeps  $\Delta\psi < 0.05 \text{ V}$  per element — inside the Scharfetter–Gummel  $V_T = kT/q$  regime.



# Bevel termination — 10-step staircase



Zoomed left bevel,  $x = -24$  to  $0 \mu\text{m}$ ,  $y = 0$  to  $15 \mu\text{m}$ .

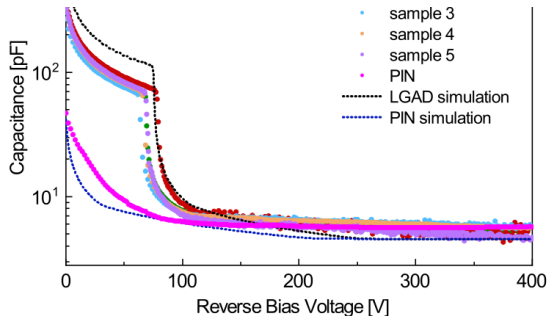
## Structure:

- 10 discrete steps per side
- Each:  $1.059 \mu\text{m}$  wide,  $0.13 \mu\text{m}$  tall
- Spans  $x = -10$  to  $-20.59 \mu\text{m}$  (left), mirrored right
- $\text{SiO}_2$  fills the stair cavities

## Physical rationale:

An abrupt gain-layer edge at  $x = \pm 4 \mu\text{m}$  would create a **field-crowding singularity**, causing premature edge breakdown. The staircase distributes the lateral field gradient — the SiC analogue of JTE / floating field rings.

# SICAR C-V target — the gain-layer step



SICAR electrical paper, Fig. 3 — C-V of PIN and LGAD.

## The signature:

- Capacitance falls steeply from  $\sim 300$  pF at 0 V
- Step / plateau at 65–80 V — gain layer fully depletes
- Slow decrease to  $\sim 30$  pF at 400 V

**PIN reference:** monotonic  $1/\sqrt{V}$  decay, **no step**.

This step is the **unique LGAD fingerprint** — the validation target for my C-V.

## CV diagnostic #1: II “ghost” conductance

**Physics.** The AC solver builds  $Y = G + j\omega C$ . Impact ionisation ( $\alpha \propto \exp(-B/E)$ ) is exponential in  $E$ . Linearising this exponential for the small-signal perturbation produces huge Jacobian derivatives  $\partial\alpha/\partial E$  that inject a spurious parallel conductance  $G$  into the imaginary part of  $Y$ , corrupting the extracted capacitance. **Symptom.** C-V shows wobbles or kinks that do not correspond to any depletion boundary. **Fix.** Disable II for C-V; enable it only for I-V (with gradqfl). Strip transport models (auger, fldmob, consrh) that decouple from electrostatics in deep reverse bias — in depletion,  $n, p \approx 0$ .

```
# C-V deck: stripped to electrostatic core
models material="4H-SiC" \
    fermidirac ni.fermi incomplete inc.two_level \
    bgn.lindefelt srh
# impact, auger, fldmob, consrh: OFF for C-V
```

## CV diagnostic #2: Robin boundary current injection

**Physics.** `con.resist` ( $\Omega \cdot \text{cm}^2$ ) converts the Dirichlet boundary (fixed  $V$ ) into a Robin boundary (mixed  $V/I$ ):

$$J_{\text{forced}} = \frac{V_{\text{applied}} - V_{\text{surface}}}{R_c}$$

With  $R_c = 1.8 \times 10^{-5}$  and  $V = 0.5 \text{ V}$ , the solver must inject  $J \approx 2.8 \times 10^4 \text{ A/cm}^2$  into a fully depleted bulk. To satisfy  $\nabla \cdot J = 0$ , it injects spurious carriers at the contact, destabilising the Poisson solve (residual  $\rightarrow 10^{19}$ ). **Symptom.** Infinite cutback loop at the first bias step. **Fix.**

Remove `con.resist` for the C-V sweep. DC leakage at low reverse bias is  $\sim 10^{-26} \text{ A}$ , so the voltage drop across any realistic  $R_c$  is numerically zero. Pure Dirichlet (ideal ohmic) is correct.

*# WRONG: Robin boundary, forced current injection*

`contact name=cathode con.resist=1.8e-5`

*# CORRECT: pure Dirichlet (omit con.resist)*

## CV diagnostic #3: 2D→3D capacitance scaling

**Physics.** A 2D TCAD simulation defaults to  $1\text{ }\mu\text{m}$  depth in  $Z$ . For my  $48\text{ }\mu\text{m}$ -wide cross-section, the simulated area ( $4.8 \times 10^{-7}\text{ cm}^2$ ) is roughly four orders of magnitude smaller than the physical SICAR pad ( $0.79\text{ mm}^2$ ). The raw capacitance comes out in fF, while the experimental baseline is  $\sim 5\text{ pF}$ . **Symptom.** Simulated C-V sits at fF-scale, three to four orders below the experimental pF-scale curve. **Fix.** Apply `width= 16458.33` to the mesh load. This integrates the displacement current over the exact 3D volume, yielding pF-scale values directly from the AC solver.

$$\text{Scaling factor} = 0.79\text{ mm}^2 / (48\text{ }\mu\text{m} \times 1\text{ }\mu\text{m}) = 16458.33.$$

## Next steps

### 1. Fix the solver

Apply the three diagnostics: strip C-V models, remove `con.resist`, apply width scaling. Target a convergent C-V sweep to 400 V.

### 2. Extract I-V and C-V curves

Run the C-V deck (impact OFF) and the I-V deck (impact ON, `gradqf1`) separately. Stitch in Python if needed.

### 3. Add traps: $Z_{1/2}$ and $EH_{6/7}$

Introduce the two dominant deep-level defects at  $10^{15} \text{ cm}^{-3}$ , with trap-assisted tunneling, for the leakage-floor deck.

### 4. Validate with SICAR

Overlay my C-V on the SICAR experimental step at  $\sim 80 \text{ V}$ , and my I-V on the SICAR leakage curve.